

TEST CASE OF THE CRON JOB HEALTH DETECTOR:

#	Test Case Method	Target Feature / Endpoint	Validation Description
1	test_health_check	/health (GET)	Asserts the API engine bootstrap is online and returns a {"status": "ok"} response with status code 200.
2	test_row_to_dict	DB row helper utility	Verifies conversion of SQL rows to standard Python dictionaries, including edge cases (e.g. returning None if input row is empty).
3	test_get_project_settings	/api/project (GET)	Verifies default dashboard parameters (dashboard name, operator, SRE descriptions) fetch cleanly.
4	test_update_project_settings	/api/project (POST)	Asserts updates to the dashboard configurations persist and propagate to the project context.
5	test_list_configured_jobs_empty	/api/jobs (GET)	Verifies that a clean system database correctly starts with zero jobs listed.
6	test_add_configured_job	/api/jobs (POST)	Asserts job configurations (Cron schedule, script file, CLI command) write and list successfully.
7	test_add_configured_job_missing_name	/api/jobs (POST) validation	Verifies input validation is active and aborts with 400 Bad Request if a job configuration name is omitted.
8	test_delete_configured_job	/api/jobs/<id> (DELETE)	Confirms registration removal logic deletes job entries and references.

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9	test_simulate_job_run	/api/jobs/simulate (POST)	Verifies random job executors start, create mock telemetry durations, and write to the run logs.
10	test_run_configured_job	/api/jobs/<id>/run (POST)	Runs a command shell wrapper manually, capturing console exits, standard streams, and operator trigger history.
11	test_get_run_by_id_and_legacy	/api/jobs/runs/<id> (GET)	Validates run detail history retrievals using both standard API parameters and legacy compatibility routing endpoints.
12	test_get_run_not_found	/api/jobs/runs/<id> (GET)	Validates bounds check error states, returning a 404 Not Found for invalid execution IDs.
13	test_prune_runs	/api/jobs/runs/prune (POST)	Confirms log pruner sweeps execution histories and resets all cron statuses back to Healthy.
14	test_anomaly_engine_rules	Anomaly Scoring Algorithm	Asserts mathematical scoring rules: assigns 0.0 for successful runs, and calculates 0.9 for processes exiting with non-zero status codes + stderr logs.
15	test_llm_narrator_helpers	Log Report Narrator	Asserts narrator builders render correct, coherent logs summary text blocks for dashboard reports and status screens.

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#	Test Case Method	Target Feature / Endpoint	Validation Description
16	test_llm_service_fallback	AI Fallback Engine	Asserts fallback SRE replies for incidents, health status, and the team members ("Gen Techies") when offline.

Test case through the command line in terminal:

```
PS D:\FINAL GEN> python -m unittest test_app.py
E
=====
ERROR: test_app (unittest.loader._FailedTest.test_app)
-----
ImportError: Failed to import test module: test_app
Traceback (most recent call last):
  File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.13_3.13.3568.0_x64__qbz5n2kfra8p0\Lib\
unittest\loader.py", line 137, in loadTestsFromName
    module = __import__(module_name)
ModuleNotFoundError: No module named 'test_app'

-----

Ran 1 test in 0.001s

FAILED (errors=1)
PS D:\FINAL GEN> python -m unittest test_app.py
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unittest\loader.py", line 137, in loadTestsFromName
```

```
    module = __import__(module_name)
```

ModuleNotFoundError: No module named 'test_app'

Ran 1 test in 0.000s

FAILED (errors=1)

PS D:\FINAL GEN> python -m unittest test_app.CronMonitorTestCase.test_health_check
test_app.CronMonitorTestCase.test_anomaly_engine_rules

>>

EE

=====

ERROR: test_app (unittest.loader._FailedTest.test_app)

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Traceback (most recent call last):

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Files\WindowsApps\PythonSoftwareFoundation.Python.3.13_3.13.3568.0_x64__qbz5n2kfra8p0\Lib\
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Traceback (most recent call last):

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Files\WindowsApps\PythonSoftwareFoundation.Python.3.13_3.13.3568.0_x64__qbz5n2kfra8p0\Lib\
unittest\loader.py", line 137, in loadTestsFromName

```
module = __import__(module_name)
```

ModuleNotFoundError: No module named 'test_app'

Ran 2 tests in 0.001s

FAILED (errors=2)

PS D:\FINAL GEN> python -m unittest discover -s cron_job/cron_job/cron_monitor -p "test_*.py"

>>

.....

Ran 16 tests in 0.477s

OK

PS D:\FINAL GEN> python -m unittest discover -s . -p "test_*.py"

>>

Ran 0 tests in 0.000s

NO TESTS RAN

PS D:\FINAL GEN> python -m unittest discover -s . -p "test_*.py"

>>

Ran 0 tests in 0.000s

NO TESTS RAN

TEST CASE OF THE CRON JOB HEALTH DETECTOR:

```
PS D:\FINAL GEN> cd cron_job/cron_job/cron_monitor
```

```
>>
```